

AnalystLab Africa Consulting
Schedule Improvement Recommendations

AI-Powered Customer Churn Prediction & Retention System: Tarrose Retail

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This report translates the findings of the Advanced Critical Path Analysis and Schedule Risk Assessment into specific actions to protect the 23-Dec-2026 completion date. It addresses critical activities, high-risk activities, resource constraints, dependency bottlenecks, and possible delays, with a concrete action against each.

1. Protect the Near-Critical/Critical Band (7.1–8.4)

Production Deployment through Sign-off has only 0–2 working days of float - smaller than any single-day slip in a source system, an approval delay, or a public holiday. This is the single highest-value place to act.

- Run a Production Deployment dry-run in staging during Week 17, while Phase 4.0 still has float, so issues surface before the real 8-day deployment window opens (addresses SR-07).
- Reserve Sponsor and Compliance Officer calendar time for the 4.6 Model Approval and 7.1 production-readiness sign-offs in advance, rather than requesting availability when the gate is reached (addresses SR-03).
- Treat any slip on 7.2 Monitoring Process or 7.3 Retraining Process as a same-day escalation to the Project Manager at 2 days of float, a one-day delay left unaddressed for 48 hours becomes a missed finish date.

2. Manage the Single-Threaded Resource Risk on the Model Chain

The Data Scientist is the only named resource across Model Development, Validation, and Explainability (4.3–4.5), all sitting directly on the critical chain, with no named backup (SR-02).

- Identify a contract data scientist or internal AnalystLab colleague who could be onboarded within 3 working days, before Week 7 rather than reactively.
- Avoid scheduling the Data Scientist onto any other AnalystLab engagement during Weeks 7–18 without Project Manager sign-off, even at partial allocation.

3. Resolve the Phase-Gate Bottleneck Explicitly

The 2.0 and 3.0 phase-gate checkpoints require every feeding activity to finish before the next phase opens, which is why a single slow task inside a gated phase can silently consume the whole phase's schedule buffer (SR-01).

- Track each gate-feeding activity individually in the weekly schedule review not the phase as a single line, so a lagging task is visible before it reaches the gate date.
- Where a gate-feeding activity is genuinely at risk, reassign float-rich resource (e.g. Data Engineer time not yet needed for the Dashboard track) to recover it rather than waiting for the gate to slip as a whole.

4. Use the Front-Loaded Float Deliberately, Not Passively

Phases 1.0–6.0 carry 25 or more working days of float each that buffer exists because of deliberate fast-tracking, not because those phases are unimportant. Left unmanaged, it can be quietly consumed by small delays with nobody noticing until it's gone.

- Set an internal 10-working-day floor on Phase 1–6 float: if any branch's float drops below 10 days, treat it as a formal schedule risk requiring Project Manager review, even though it is technically still non-critical.
- Confirm Marketing & Customer Service Team availability windows in writing for Weeks 11–17 during Phase 2.0 resource planning, rather than assuming the 'light involvement' noted in the Charter will materialize on schedule (addresses SR-06).
- Pre-agree with the IT Lead that no Tarrose Retail system changes or maintenance windows fall within Weeks 3–6, protecting the Data Assessment activities that gate the entire back half of the project (addresses SR-04).

5. Guard Against Late Scope Change on the Critical Chain

A change to the 75% accuracy target or churn-reduction criteria after Model Development starts would force rework on activities with little to no float (SR-05).

- Enforce the existing Charter change-control rule without exception: any change to the accuracy target requires formal Sponsor sign-off before action, not informal agreement in a working session.

- If a target change is approved, re-run the Critical Path Analysis immediately before committing to a revised date, so the Sponsor is deciding with the real schedule impact already known.

Summary

The schedule's real vulnerability is concentrated in four weeks, not twenty: Production Deployment through Sign-off (7.1–8.4). Protecting that band through a deployment dry-run, pre-booked approvals, and a same-day escalation habit will do more for the 23-Dec-2026 date than any amount of attention spent on the float-rich front two-thirds of the project.